

Reflection Report

Reflection on Week 1: Business Understanding & Data Exploration

This project was my first assignment as a Data Science Intern at AnalystLab Africa, and it gave me a practical introduction to how data science is used to solve real business problems. Before this assignment, I understood the technical concepts of data analysis, but this project helped me see how those skills can be applied to answer important business questions.

One of the most valuable lessons I learned was that data science is not only about writing code or creating charts. It starts with understanding the business problem. In this case, the goal was to investigate employee attrition and identify possible factors that influence employees to leave the company. This helped me appreciate the importance of connecting data analysis with business decision-making.

During the project, I worked with the dataset by inspecting its structure, checking for missing values and duplicates, exploring descriptive statistics, and creating different visualizations. Interpreting the results was more challenging than creating the charts themselves. I learned that every graph should tell a meaningful story and provide insights that can help the business make better decisions.

One challenge I faced was deciding which visualizations best answered the business questions and how to interpret them correctly. I realized that it is important not to jump to conclusions but to explain findings carefully based on the available data.

This assignment also strengthened my skills in Python, Pandas, Matplotlib, and Seaborn. I became more confident in cleaning data, exploring datasets, and communicating findings in a clear and professional manner. I also improved my ability to organize a data analysis project from business understanding to final insights.

Overall, this project has increased my interest in data science and motivated me to continue improving my analytical and visualization skills. In future projects, I would like to perform more advanced statistical analysis and build predictive machine learning models to identify employees who may be at risk of leaving. I look forward to applying the knowledge gained from this assignment to more complex real-world business problems throughout my internship.